

6. Alumco manufactures aluminum sheets and aluminum bars. The maximum production capacity is estimated at either 800 sheets or 600 bars per day. The maximum daily demand is 550 sheets and 580 bars. The profit per ton is \$40 per sheet and \$35 per bar. Determine the optimal daily production mix.

x_1 = # of sheets produced in one day.

x_2 = # of bars.

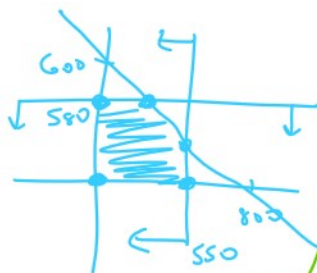
$$\max z = 40x_1 + 35x_2$$

$$\frac{x_1}{800} + \frac{x_2}{600} \leq 1$$

$$x_1 \leq 550$$

$$x_2 \leq 580$$

$$x_1, x_2 \geq 0$$



$$\frac{400}{800} = \frac{1}{2} = 50\%$$

$$50\% \times 100 \rightarrow 1$$

$$\frac{100}{100}$$

↓
 $x_1^* = 550, x_2^* = 187.13$

$$z^* = \$28549.40$$

Verify it! ▽

4. John must work at least 20 hours a week to supplement his income while attending school. He has the opportunity to work in two retail stores. In store 1, he can work between 5 and 12 hours a week, and in store 2 he is allowed between 6 and 10 hours. Both stores pay the same hourly wage. In deciding how many hours to work in each store, John wants to base his decision on work stress. Based on interviews with present employees, John estimates that, on an ascending scale of 1 to 10, the stress factors are 8 and 6 at stores 1 and 2, respectively. Because stress mounts by the hour, he assumes that the total stress for each store at the end of the week is proportional to the number of hours he works in the store. How many hours should John work in each store?

x_1 = # of hours in store 1.

x_2 = # of hours in store 2.

$$\min z = 8x_1 + 6x_2$$

$$x_1 + x_2 \geq 20$$

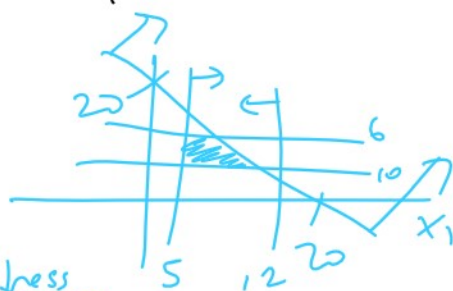
$$5 \leq x_1 \leq 12 \Rightarrow$$

$$6 \leq x_2 \leq 10$$

$$x_1, x_2 \geq 0$$

$$x_1 \leq 12$$

$$x_1 \geq 5$$



$$x_1^* = 10, x_2^* = 10$$

$$z^* = 140 \text{ is best answer.}$$

$$\text{Maximize } z = 2x_1 + 3x_2$$

$$\text{subject to } 2x_1 + x_2 \leq 4 \rightarrow (1) \Rightarrow "="$$

$$x_1 + 2x_2 \leq 5 \rightarrow (2)$$

$$x_1, x_2 \geq 0$$

$$(1) \Rightarrow 2x_1 + x_2 + S_1 = 4, S_1 \geq 0$$

$$(2) \Rightarrow x_1 + 2x_2 + S_2 = 5, S_2 \geq 0.$$

$$x_1, x_2 \geq 0.$$

$$4 \text{ variables} \Rightarrow x_1, x_2, S_1, S_2 \Rightarrow n = 4$$

$$2 \text{ equations.} \Rightarrow m = 2.$$

$$n - m = 4 - 2 = 2 \text{ variables set to zero.}$$

$$\binom{n}{m} = \binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{2 \cdot 4 \cdot 3}{2} = 6. \text{ different combinations.}$$

Variables
selected
to set to zero

Variables
that are
found.

$$n - m = 4 - 2 = 2 \quad S_1 = 4, S_2 = 5$$

feasible
infeasible

$$z = 2x_1 + 3x_2$$

$$z = 0$$

to set to zero

- ① $x_1 = 0, x_2 = 0, s_1 = 4, s_2 = 5$
~~② $x_1 = 0, s_1 = 0, x_2 = 4, s_2 = -3$~~
~~③ $x_1 = 0, s_2 = 0, x_2 = 5/2, s_1 = 3/2$~~
~~④ $x_2 = 0, s_1 = 0, x_1 = 2, s_2 = 3$~~
~~⑤ $x_2 = 0, s_2 = 0, x_1 = 5, s_1 = -6$~~
 ⑥ $s_1 = 0, s_2 = 0, x_1 = 1, x_2 = 2$

feasible
 infeasible
 feasible
 feasible
 infeasible
 feasible

$z = 0$
 $z = 3 \cdot \frac{5}{2} = \frac{15}{2}$
 $z = 4$
 $z = 8$

optimal.

$$\begin{array}{l} 2x_1 + x_2 + s_1 = 4 \\ x_1 + 2x_2 + s_2 = 5 \end{array} \quad \left. \begin{array}{l} \text{if } x_1 = x_2 = 0 \\ s_1 = 4, \\ s_2 = 5. \end{array} \right\}$$

if $x_1 = 0, s_1 = 0, x_2 = 4, s_2 = 5 - 8 = -3$.

if $s_1 = s_2 = 0,$

$$\begin{array}{rcl} 2x_1 + x_2 & = & 4 \\ -2/ & x_1 + 2x_2 & = 5 \\ \hline -2x_1 - 4x_2 & = & -10 \\ 2x_1 + x_2 & = & 4 \\ \hline \end{array}$$

$$-3x_2 = -6$$

$$\begin{array}{l} x_2 = 2 \\ x_1 = 1 \end{array}$$